THE BRIDGE INDEX AND THE NONORIENTABLE FOUR-GENUS OF KNOTS WITH AT MOST THIRTEEN CROSSINGS

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Abstract. We determine two invariants for knots that the KnotInfo database leaves open. KnotInfo lists the bridge index of 536 prime knots with thirteen crossings as either 3 or 4; we prove that 243 of them have bridge index and meridional rank 4, so the meridional rank conjecture holds for them. The lower bound comes from representations of the knot group in $\mathrm{GL}_4(\mathbb{F}_q)$ that send a meridian to a transvection or a reflection and fix no nonzero vector. KnotInfo leaves the smooth nonorientable four-genus open for 3525 prime knots with eleven to thirteen crossings; we determine it for 2537 of them and narrow the interval for 376 more. Upper bounds come from pinch moves, lower bounds from the Goeritz-lattice obstructions of Jabuka and Kelly evaluated by an exhaustive search for lattice embeddings. Every bound is a certificate checked by a program independent of the search that found it.

1. Introduction

The KnotInfo database [17] records, for each prime knot with at most thirteen crossings, the values of several dozen invariants, and an interval where the value is not known. This paper closes many of these intervals for two invariants, the bridge index and the smooth nonorientable four-genus, both in the version 2026.10.5 of the database. The two parts are independent and use different methods; they share the source of the diagrams, the form of the results, a list of certificates, one per knot, each checked by a short program, and the treatment of the knots that remain open.

The bridge index. Let $K \subset S^3$ be a knot, $G = \pi_1(S^3 \setminus K)$ its group and $m \in G$ a meridian. The *bridge index* $b(K)$ is the least number of local maxima of a height function on a knot isotopic to K. The *meridional rank* $\mu(K)$ is the least number of meridians, that is conjugates of $m^{\pm 1}$, that generate G. The overpasses of a b-bridge presentation give b meridians generating G, so $\mu(K) \leq b(K)$. Cappell and Shaneson asked whether equality always holds [15, Problem 1.11]; this is the meridional rank conjecture.

Blair, Kjuchukova, Velazquez and Villanueva defined the Wirtinger number $\omega(D)$ of a diagram D and proved that its minimum over the diagrams of K is $b(K)$ [3]. In particular

$$b(K) \leq \omega(D) \quad \text{for every diagram } D \text{ of } K. \tag{1.1}$$

Blair, Kjuchukova and Morrison bounded $\mu(K)$ from below by surjections of G onto Coxeter groups sending meridians to reflections, and so computed $b(K) = \mu(K)$ for 595,515 knots up to sixteen crossings [1]. KnotInfo records the bridge index of thirteen-crossing knots from these bounds and from the plain sphere number [2]. In the version 2026.10.5, 536 knots with thirteen crossings have the entry $[3, 4]$.

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Theorem 1.1. *For each of the 243 knots K listed in Appendix A,*

$$b(K) = \mu(K) = 4.$$

In particular the meridional rank conjecture holds for these knots.

The upper bound is (1.1) applied to the KnotInfo diagram of K, whose Wirtinger number is 4. The lower bound is Lemma 2.1, applied to a representation $\rho\colon G \to \mathrm{GL}_4(\mathbb{F}_q)$ with $q \in \{3, 5\}$ that sends the meridian to a transvection of $\mathrm{Sp}_4(\mathbb{F}_q)$ or to a reflection of an orthogonal group $\mathrm{O}_4(\mathbb{F}_5)$. Reflections in Coxeter groups are pseudo-reflections too; the transvections, of order 3 and 5, are what settles most of these knots. Table 1 gives the counts.

group	image of a meridian	knots
$\mathrm{O}_4(\mathbb{F}_5)$	reflections	15
$\mathrm{Sp}_4(\mathbb{F}_3)$	transvections	67
$\mathrm{Sp}_4(\mathbb{F}_5)$	transvections	161

TABLE 1. The certificates of Theorem 1.1, by the group used.

The nonorientable four-genus. Every knot $K \subset S^3 = \partial B^4$ bounds a smooth surface in B^4; if K is not slice, it bounds nonorientable ones as well. The *nonorientable four-genus* $\gamma_4(K)$ is the least first Betti number of a smooth compact nonorientable surface $F \subset B^4$ with $\partial F = K$; for slice knots one sets $\gamma_4(K) = 1$. It was studied by Murakami and Yasuhara [18] and by Gilmer and Livingston [10]. Its values on small knots were computed by Jabuka and Kelly for eight and nine crossings [14], by Ghanbarian for ten [9] and by Fairchild for nonalternating knots with eleven [8]. Lee and Sabloff [16] extended the computation to fourteen crossings with an algorithmic search for Möbius bands in the style of Dunfield and Gong [7], together with the Arf invariant obstruction of Yasuhara [20] and a linking form obstruction of Murakami and Yasuhara [18]. KnotInfo incorporates these data. In its version 2026.10.5, 3525 knots with eleven to thirteen crossings have an interval $[a, b]$ with $a < b$ as the entry for γ_4.

Theorem 1.2. *The value $\gamma_4(K)$ is determined for 2537 of these knots: 520 have $\gamma_4(K) = 1$, 1995 have $\gamma_4(K) = 2$ and 22 have $\gamma_4(K) = 3$. For 2177 of them a pinch move brings the upper end of the KnotInfo interval down to its lower end; for 163 the lower bound of Theorem 7.1 raises the lower end to the upper end; for 197 the two meet in between. For 376 further knots the interval is narrowed.*

The knots are listed in Appendix C, and Table 3 gives the counts by crossing number and by KnotInfo interval. Part I (Sections 2 to 4) proves Theorem 1.1, Part II (Sections 5 to 8) proves Theorem 1.2. Both parts end with what the methods do not reach.

Part I. The bridge index

2. A lower bound from pseudo-reflections

A *pseudo-reflection* of a vector space V is an element $g \in \mathrm{GL}(V)$ with $\operatorname{rank}(g - 1) = 1$. It fixes the hyperplane $\ker(g - 1)$ pointwise.

Lemma 2.1. *Let V be a vector space of dimension n over a field and let $\rho\colon G \to \mathrm{GL}(V)$ be a homomorphism such that $\rho(m)$ is a pseudo-reflection. Let V^G be the subspace of vectors fixed by $\rho(G)$. Then*

$$\mu(K) \;\geq\; n - \dim V^G.$$

Proof. Let $m_1, \dots, m_k$ be meridians generating G, with $m_i = g_i m^{\epsilon_i} g_i^{-1}$ and $\epsilon_i = \pm 1$. Since $\rho(m)^{-1} - 1 = -\rho(m)^{-1}(\rho(m) - 1)$, each $\rho(m_i)$ is a pseudo-reflection, and $H_i = \ker(\rho(m_i) - 1)$ is a hyperplane. The intersection $H_1 \cap \cdots \cap H_k$ has dimension at least $n - k$ and is fixed by every $\rho(m_i)$, hence by $\rho(G)$. So $n - k \le \dim V^G$. □

Corollary 2.2. *If* $\dim V = 4$, $\rho(m)$ *is a pseudo-reflection and* $V^G = 0$, *then* $b(K) \ge \mu(K) \ge 4$.

In a diagram with arcs $x_1, \dots, x_r$, the Wirtinger presentation makes such a ρ an assignment of matrices $X_i = \rho(x_i)$ to the arcs that satisfies $X_c = X_o^{\pm 1} X_a X_o^{\mp 1}$ at each crossing, where x_o is the overarc and x_a, x_c are the underarcs. Since the X_i generate $\rho(G)$,

$$V^G = \bigcap_{i=1}^{r} \ker(X_i - 1).$$

Both conditions are finite computations once the X_i are written down.

Transvections. Let ω be a nondegenerate alternating form on $V = \mathbb{F}_q^4$. For $v \neq 0$ the transvection $t_v(x) = x + \omega(x, v)\, v$ lies in $\mathrm{Sp}(V, \omega)$, $t_v - 1$ has rank one, $t_{-v} = t_v$, and $g\, t_v\, g^{-1} = t_{gv}$ for $g \in \mathrm{Sp}(V, \omega)$. A representation with $X_i = t_{v_i}$ is therefore the same thing as a coloring of the arcs by vectors with

$$v_c = \pm\, t_{v_o}^{\pm 1}(v_a)$$

at every crossing, and $\ker(X_i - 1) = v_i^{\perp}$. So $V^G = (\mathrm{span}\{v_i\})^{\perp}$, which is 0 exactly when the v_i span V.

Reflections. Let B be a nondegenerate symmetric bilinear form on $\mathbb{F}_q^4$, q odd. For $B(v, v) \neq 0$ the reflection $r_v(x) = x - 2\frac{B(x,v)}{B(v,v)}\, v$ satisfies $g\, r_v\, g^{-1} = r_{gv}$ for $g \in \mathrm{O}(B)$, and again $V^G = (\mathrm{span}\{v_i\})^{\perp}$.

3. The certificates for the bridge index

For each knot of Appendix A the file `data/certificates.json` contains:

- the PD code of K from KnotInfo, with edges numbered $1, \dots, 2n$ along the orientation and $[a, b, c, d]$ the edges at a crossing counterclockwise from the incoming underedge a;
- a prime q and a 4×4 matrix over $\mathbb{F}_q$ for every edge;
- four edges on distinct arcs, the seeds of a Wirtinger coloring.

The program `verify/verify.py` uses only the Python standard library. For every knot it checks:

(1) the matrices are constant along each arc, where an arc is a maximal run of edges between two undercrossings;
(2) at each crossing $[a, b, c, d]$ the relation $X_c = X_b^{-1} X_a X_b$ holds if the overstrand runs from b to d, and $X_c = X_b X_a X_b^{-1}$ if it runs from d to b (changing this convention at every crossing at once replaces K by its mirror image, which has the same b and μ);
(3) each $X_i - 1$ has rank 1, and the matrices $X_i - 1$ stacked together have rank 4, so that $V^G = 0$;
(4) starting from the four seed arcs, the coloring moves of [3] (if the overarc and one underarc at a crossing are colored, color the other underarc) color every arc, so $\omega(D) \le 4$.

By Corollary 2.2 and (1.1), items (1)–(4) prove $b(K) = \mu(K) = 4$. With the option `--knotinfo` the program also checks that each PD code is the one in KnotInfo and that KnotInfo lists the bridge index as $[3, 4]$. All 243 certificates are checked in a few seconds. The program rejects altered certificates: a changed matrix entry, a seed set on three arcs, one crossing with its overstrand reversed, or identity matrices.

Remark 3.1. The certificates were found by a search, which plays no part in the proof. If the four seed vectors do not span V, every arc vector lies in their span, so a certificate must have seeds forming a basis. A coloring is then determined by the Gram matrix of the seeds: six entries of an alternating matrix, or six off-diagonal entries of a symmetric matrix with a fixed diagonal. The search runs over all such matrices and propagates the colors from the seeds through the crossings. The Wirtinger relations are then checked at the remaining crossings.

4. The knots whose bridge index remains open

For the other 293 knots with entry $[3, 4]$ (Appendix B) we found neither bound. The exhaustive searches described above found no representation satisfying Corollary 2.2 with meridians sent to transvections of $\mathrm{Sp}_4(\mathbb{F}_q)$ for $q \in \{3, 4, 5, 7, 8, 9, 11\}$, to reflections of $\mathrm{O}_4(\mathbb{F}_q)$ for $q \in \{3, 5, 7\}$ (both isometry types), or to pseudo-reflections of $\mathrm{GL}_4(\mathbb{F}_2)$ and $\mathrm{GL}_4(\mathbb{F}_3)$ with eigenvalue 1 or -1. Random Reidemeister moves (SnapPy [5]) produced no diagram of Wirtinger number 3; 300 diagrams were tried for each knot. Abelian colorings give nothing either: for every prime $p < 110$ and every irreducible factor g of the Alexander polynomial modulo p, the colorings of the diagram by the Alexander quandle $\mathbb{F}_p[t]/(g)$, $a \triangleright b = ta + (1-t)b$, form a space of dimension at most 3, while dimension 4 would give $\mu(K) \geq 4$. Nor is any of these knots the closure of a 3-braid $\Delta^{2d}w$ in Murasugi's normal form with $|d| \leq 4$ and w either $\sigma_1^{-p_1}\sigma_2^{q_1}\cdots\sigma_1^{-p_s}\sigma_2^{q_s}$ of length at most $17 - 4\max(|d|-1, 0)$ or σ_2^m with $|m| \leq 20$ (53,309 braids, identified by SnapPy), which would give $b(K) \leq 3$. A diagram of Wirtinger number 3 would give $b(K) = \mu(K) = 3$, and a representation as in Corollary 2.2 over another field, or a Coxeter quotient of rank 4, would give $b(K) = \mu(K) = 4$.

Part II. The nonorientable four-genus

5. Pinch moves

Definition 5.1. Let D be a diagram of a knot K and c a crossing of D. Of the two smoothings of c, one respects the orientation of K and gives a two-component link; the other gives a knot K'. We say K' is obtained from K by a *pinch move* at c.

Lemma 5.2. *If K' is obtained from K by a pinch move, then $\gamma_4(K) \leq 1$ when K' is slice, and $\gamma_4(K) \leq \gamma_4(K') + 1$ in general.*

Proof. The pinch move is a band move, so there is a cobordism $\Sigma \subset S^3 \times [0,1]$ from K to K' formed by $K \times [0,1]$ and one band. Since K' is connected, the band does not respect the orientation of K, and Σ is a Möbius band with an open disk removed. If K' bounds a disk Δ in B^4, then $\Sigma \cup \Delta$ is a Möbius band bounded by K, so $\gamma_4(K) \leq 1$. If K' bounds a surface F with $b_1(F) = \gamma_4(K')$, then $\Sigma \cup F$ is a nonorientable surface bounded by K with $b_1 = b_1(F) + 1$. This is the bound used by Jabuka and Kelly [14]. □

A knot that is not slice has $\gamma_4 \geq 1$. For the knots whose KnotInfo interval starts at 2, the lower bound is the obstruction to bounding a Möbius band recorded by KnotInfo from Lee and Sabloff [16]: either Yasuhara's condition $4\,\mathrm{Arf}(K) - \sigma(K) \in \{0, \pm 2\} \pmod 8$ fails [20], or the linking form obstruction of Murakami and Yasuhara [18] applies. The further lower bounds are those of Section 7. Theorem 1.2 therefore rests on these obstructions, on the slice status and four-genus values recorded in KnotInfo for the targets, on the certificates of Section 6 and on the lattice computations of Section 7.

6. The pinch certificates and their verification

A certificate for a knot K consists of a planar diagram code D, a crossing c of D, a choice of smoothing, and a target: either the unknot or a knot T of KnotInfo. It asserts that

(1) D is a diagram of K;
(2) the chosen smoothing of c gives a one-component diagram D';
(3) D' represents T or its mirror image;
(4) the bound $\beta = 1$ if T is slice or the unknot, and $\beta = \gamma_4(T) + 1$ otherwise, is smaller than the upper end of the KnotInfo interval of K.

Since γ_4 and sliceness are invariant under mirror image, (1) to (4) and Lemma 5.2 give $\gamma_4(K) \leq \beta$.

The certificates were found with SnapPy [5]: for every open knot we applied every pinch move to the KnotInfo diagram and to up to twenty-five diagrams obtained from it by random Reidemeister moves, and identified the resulting knots by hyperbolic structure. A candidate enters the file only if the verifier below accepts it; when it did not, the search went on with other crossings and other diagrams of the same knot. A second pass used the knots settled in the first pass as targets; the number of certificates with such a target is 2, and the verifier checks them after the certificates of their targets. A last pass applied every pinch move to twenty diagrams of each knot left open.

The verifier `crosscap4d/verify/verify.py` does not use SnapPy. It reads the KnotInfo database and uses Regina [4] for the topology. For (1) and (3) it proves that two diagrams represent the same knot up to mirror image in one of two ways.

- *Isotopy.* Both diagrams are simplified by Reidemeister moves to the same number of crossings, and the closures of the two minimal diagrams under Reidemeister moves that add at most one crossing share a diagram. Diagrams are compared by Regina's canonical signatures, taken up to reflection, rotation and reversal.
- *Complements.* The triangulated complements of the two diagrams have combinatorially isomorphic triangulations, possibly after Pachner moves that add at most one tetrahedron, or one of the triangulations obtained from forty simplifications of each complement under fixed seeds of Regina's random engine, together with the triangulations one Pachner move away from the smallest of them, occurs for both diagrams. Each of these triangulations is a triangulation of the complement it came from, so a common one shows that the complements are homeomorphic, and by the theorem of Gordon and Luecke [12] the knots agree up to mirror image.

Both tests are exact: a positive answer is a proof. Regina's simplification is randomised, so the verifier makes up to four attempts, each from a fixed seed, and gives each certificate two minutes. For the unknot the verifier uses Regina's recognition of the solid torus. Item (2) is checked by following the strands of D', and (4) by reading KnotInfo.

The verifier accepts all 2728 certificates. It rejects certificates corrupted in four ways, on a sample of twenty: the other smoothing of the crossing, the next crossing, a wrong target, and a bound lowered by one (Table 2). The seven modified certificates with the next crossing that it accepts are valid certificates: that crossing gives the same target.

7. Lower bounds from definite lattices

Let D be a diagram of a knot K with a checkerboard colouring. Following Gordon and Litherland [11], give each crossing c the weight $\eta(c) = \pm 1$ according to the position of the shaded regions at c, let $X_0, \ldots, X_n$ be the unshaded regions, and let G be the $n \times n$ matrix with $g_{ij} = -\sum \eta(c)$ over the crossings meeting X_i and X_j for $i \neq j$, and $g_{ii} = -\sum_{k \neq i} g_{ik}$ with k running from 0 to n. This is the Goeritz matrix; $\det G = \pm \det K$, and with $\mu = \sum \eta(c)$ over

corruption	rejected
other smoothing of the crossing	20 of 20
next crossing of the diagram	13 of 20
wrong target ($12a_1$)	20 of 20
bound lowered by one	20 of 20

TABLE 2. Corrupted pinch certificates rejected by the verifier.

the crossings of type II for the shaded surface,

$$\sigma(K) = \operatorname{sign}(G) - \mu. \tag{7.1}$$

The form G is the intersection form of the double cover of B^4 branched along the shaded surface pushed into B^4 [11]. If D is reduced and alternating, the forms G_+ and G_- of the two colourings are positive and negative definite.

Theorem 7.1 (Jabuka and Kelly [14, Theorems 2.10–2.12])**.** *Let K be a knot with a reduced alternating diagram, let $P_+ = G_+$ and $P_- = -G_-$, of ranks $r_\pm$, and let $s \equiv \sigma(K) + 4\operatorname{Arf}(K) \pmod 8$. Call $\ell \in \mathbb{N}$ admissible if $\ell \mid \det K$ and $\det K/\ell$ is a square.*

1. *If $s = 2$ and $P_- \oplus [\ell]$ embeds in $\mathbb{Z}^{r_-+1}$ for no admissible ℓ, or $s = 6$ and $P_+ \oplus [\ell]$ embeds in $\mathbb{Z}^{r_++1}$ for no admissible ℓ, then $\gamma_4(K) \geq 2$.*
2. *If $s = 0$ and neither $P_+ \oplus [\ell]$ embeds in $\mathbb{Z}^{r_++1}$ nor $P_- \oplus [\ell]$ in $\mathbb{Z}^{r_-+1}$, for any admissible ℓ, then $\gamma_4(K) \geq 2$.*
3. *If $s = 4$ and neither P_+ embeds in $\mathbb{Z}^{r_++2}$ nor P_- in $\mathbb{Z}^{r_-+2}$, then $\gamma_4(K) \geq 3$.*

Here $\mathbb{Z}^N$ carries the standard positive definite form.

We recall the argument. If K bounds a Möbius band $\Sigma \subset B^4$, the double branched cover $W(\Sigma)$ has $b_1 = 0$ [14] and $b_2 = 1$, its form is $[\pm\ell]$ with ℓ admissible by a theorem of Gilmer and of Owens and Strle [14, Corollary 2.8], and the congruence of Gilmer and Livingston [10] fixes the sign of ℓ by s. Gluing $W(\Sigma)$ with reversed orientation to the double cover bounded by the shaded surface whose Goeritz form has the opposite sign gives a closed definite smooth four-manifold whose form contains $G \oplus [\mp\ell]$ in full rank. By Donaldson's theorem [6] that form is diagonal; changing the sign of the whole form gives (1) and (2). In (3) a surface with $b_1 = 2$ would give a definite $W(\Sigma)$ of rank two, of either sign, and the same gluing embeds P_+ or P_- in a diagonal lattice of rank two more. In (1) the statement of [14] names the target lattice as ε-definite; the argument above, which is theirs, produces the lattice of the sign of G, and it is in this form that we apply it.

Since γ_4 is invariant under mirror image, and mirroring changes the sign of every weight $\eta(c)$, of G, of μ and of σ, it does not matter which of the two weight conventions is called positive, as long as σ is computed from (7.1) with the same weights.

Lemma 7.2. *Let P be a positive definite lattice of rank r and $\ell \in \mathbb{N}$. Then $P \oplus [\ell]$ embeds in $\mathbb{Z}^{r+1}$ if and only if there are an embedding $\phi\colon P \to \mathbb{Z}^{r+1}$ and an integer $k \geq 1$ with $\ell = k^2|w|^2$, where w is a primitive vector spanning the orthogonal complement of $\phi(P)$.*

Proof. The orthogonal complement of a sublattice of rank r in $\mathbb{Z}^{r+1}$ is $\mathbb{Z}w$ for a primitive w, and a vector of norm ℓ orthogonal to $\phi(P)$ is kw with $k^2|w|^2 = \ell$. □

Lemma 7.3. *Let P be a positive definite Gram matrix of rank r and $N \geq r$. Every embedding of P in $\mathbb{Z}^N$ is carried by a signed permutation of the coordinates to one found by the following search. The basis vectors $e_1, \dots, e_r$ are placed in turn; if the vectors already placed are supported on the first t coordinates, the next vector is $(x, y, 0)$ with $x \in \mathbb{Z}^t$ arbitrary, subject to the norm*

and the inner products with the vectors already placed, and y a nonincreasing sequence of positive integers.

Proof. By induction on the number of vectors placed. Given an embedding $v_1, \dots, v_r$ whose first $i-1$ vectors are of the stated form, supported on the first t coordinates, the signed permutations of the coordinates beyond t fix $v_1, \dots, v_{i-1}$; one of them makes the entries of v_i beyond t nonnegative and nonincreasing, with its nonzero entries first, and it does not change the shape of the vectors already placed. □

We take the KnotInfo diagram of each open alternating knot, compute both Goeritz forms and σ from (7.1), check that the two colourings give the same σ, that $|\sigma|$ agrees with KnotInfo and that the forms are definite, and run the search of Lemma 7.3 to the end: for (1) and (2) it lists the norms $|w|^2$ of Lemma 7.2 over all embeddings, for (3) it decides whether an embedding exists. Of the 3525 knots, the hypotheses of Theorem 7.1 hold and raise the KnotInfo lower end for 382: 360 reach $\gamma_4 \geq 2$ (314 by (1) and 46 by (2)) and 22 reach $\gamma_4 \geq 3$ by (3). The data are in `crosscap4d/data/lower_bounds.json`.

The program `crosscap4d/lower/verify_lower.py` recomputes every entry with the basis order of the search shuffled under a fixed seed, and runs three tests of the conventions. On the alternating knots with at most eleven crossings whose γ_4 KnotInfo records, the bound never exceeds the recorded value, while the same program with the sign of σ reversed contradicts the recorded value for more than two hundred of them. On the alternating knots that carry a pinch certificate, the bound never exceeds the upper bound of that certificate.

crossings	KnotInfo	knots	settled	narrowed	open
11	[1, 2]	14	8	0	6
11	[1, 3]	9	8	1	0
11	[2, 3]	36	33	0	3
12	[1, 2]	87	43	0	44
12	[1, 3]	87	49	35	3
12	[2, 3]	278	265	0	13
12	[2, 4]	3	3	0	0
13	[1, 2]	890	431	0	459
13	[1, 3]	677	318	338	21
13	[1, 4]	3	1	2	0
13	[1, 5]	1	0	0	1
13	[2, 3]	1437	1375	0	62
13	[2, 4]	3	3	0	0
all		3525	2537	376	612

TABLE 3. Open entries of γ_4 in KnotInfo 2026.10.5 and what the certificates give. "Settled": the value is now known; "lowered": the upper bound is lower.

8. The knots whose nonorientable four-genus remains open

The remaining 988 knots, with their narrowed intervals, are listed in the file `still_open.json` of the folder `crosscap4d/data`. For a knot with interval $[1, 2]$ that is not settled, no pinch move we tried gives a slice knot, and Theorem 7.1 does not apply or does not obstruct. Theorem 7.1 needs a definite diagram, so it says nothing about the nonalternating knots.

The bound $|\upsilon(K) - \sigma(K)/2| \leq \gamma_4(K)$ of Ozsváth, Stipsicz and Szabó [19], with $\upsilon = \Upsilon_K(1)$, does not raise any of these lower ends where we could evaluate it. For the open knots with Hom's invariant $\varepsilon(K) = 0$ the complex $CFK^\infty(K)$ is ε-equivalent to that of the unknot [13], so $\upsilon(K) = 0$ and the bound is $|\sigma(K)|/2$, which never exceeds the lower end of the KnotInfo interval. For the others, the knot Floer complexes we could compute are complexes over $\mathbb{F}[U, V]/(UV)$, which do not determine $\Upsilon_K(1)$.

Remark 8.1. The 520 Möbius bands found here are not in the data of Lee and Sabloff [16], whose search follows Dunfield and Gong [7] and runs over a finite family of bands chosen in advance. The knots settled at $\gamma_4 = 2$ were outside their aim, which was Möbius bands; for them the target of the pinch move is a knot known to bound a Möbius band but not a disk.

Data availability

The certificates, the verifiers and the search programs for both parts are in the repository `https://github.com/creelie/conjecture-to-choose`, release 1.3.0, `https://github.com/creelie/conjecture-to-choose/releases/tag/v1.3.0`: Part I in the folders `data`, `verify` and `search`, Part II in the folder `crosscap4d`.

Use of AI tools

The searches, the verification programs and drafts of the text were produced with Claude Code (Anthropic) under the direction of the author, who is responsible for the content.

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Appendix A. The knots of Theorem 1.1

$13a_{69}$, $13a_{70}$, $13a_{72}$, $13a_{73}$, $13a_{74}$, $13a_{76}$, $13a_{79}$, $13a_{96}$, $13a_{104}$, $13a_{106}$, $13a_{107}$, $13a_{108}$, $13a_{156}$, $13a_{157}$, $13a_{160}$, $13a_{171}$, $13a_{235}$, $13a_{343}$, $13a_{353}$, $13a_{354}$, $13a_{355}$, $13a_{358}$, $13a_{359}$, $13a_{360}$, $13a_{388}$, $13a_{390}$, $13a_{396}$, $13a_{399}$, $13a_{400}$, $13a_{402}$, $13a_{403}$, $13a_{408}$, $13a_{413}$, $13a_{414}$, $13a_{505}$, $13a_{524}$, $13a_{525}$, $13a_{526}$, $13a_{527}$, $13a_{528}$, $13a_{529}$, $13a_{535}$, $13a_{536}$, $13a_{541}$, $13a_{542}$, $13a_{543}$, $13a_{544}$, $13a_{545}$, $13a_{554}$, $13a_{555}$, $13a_{557}$, $13a_{558}$, $13a_{596}$, $13a_{597}$, $13a_{601}$, $13a_{621}$, $13a_{630}$, $13a_{631}$, $13a_{632}$, $13a_{636}$, $13a_{637}$, $13a_{657}$, $13a_{664}$, $13a_{665}$, $13a_{815}$, $13a_{867}$, $13a_{1392}$, $13a_{1538}$, $13a_{1995}$, $13a_{2006}$, $13a_{2008}$, $13a_{2613}$, $13a_{2614}$, $13a_{2615}$, $13a_{2616}$, $13a_{2617}$, $13a_{2619}$, $13a_{2620}$, $13a_{2800}$, $13a_{2801}$, $13a_{3038}$, $13a_{3040}$, $13a_{3107}$, $13a_{3144}$, $13a_{3146}$, $13a_{3179}$, $13a_{3257}$, $13a_{3333}$, $13a_{3334}$, $13a_{3343}$, $13a_{3344}$, $13a_{3376}$, $13a_{3377}$, $13a_{3378}$, $13a_{3380}$, $13a_{3381}$, $13n_{160}$, $13n_{161}$, $13n_{162}$, $13n_{163}$, $13n_{164}$, $13n_{165}$, $13n_{166}$, $13n_{167}$, $13n_{168}$, $13n_{169}$, $13n_{173}$, $13n_{174}$, $13n_{175}$, $13n_{176}$, $13n_{177}$, $13n_{178}$, $13n_{179}$, $13n_{180}$, $13n_{181}$, $13n_{182}$, $13n_{211}$, $13n_{429}$, $13n_{430}$, $13n_{431}$, $13n_{432}$, $13n_{433}$, $13n_{434}$, $13n_{435}$, $13n_{440}$, $13n_{441}$, $13n_{442}$, $13n_{449}$, $13n_{467}$, $13n_{601}$, $13n_{609}$, $13n_{610}$, $13n_{929}$, $13n_{930}$, $13n_{931}$, $13n_{932}$, $13n_{933}$, $13n_{957}$, $13n_{958}$, $13n_{959}$, $13n_{960}$, $13n_{961}$, $13n_{962}$, $13n_{963}$, $13n_{964}$, $13n_{965}$, $13n_{966}$, $13n_{967}$, $13n_{968}$, $13n_{977}$, $13n_{978}$, $13n_{979}$, $13n_{980}$, $13n_{981}$, $13n_{982}$, $13n_{983}$, $13n_{1042}$, $13n_{1058}$, $13n_{1059}$, $13n_{1060}$, $13n_{1061}$, $13n_{1062}$, $13n_{1063}$, $13n_{1064}$, $13n_{1072}$, $13n_{1073}$, $13n_{1074}$, $13n_{1075}$, $13n_{1076}$, $13n_{1077}$, $13n_{1078}$, $13n_{1082}$, $13n_{1083}$, $13n_{1084}$, $13n_{1103}$, $13n_{1104}$, $13n_{1105}$, $13n_{1106}$, $13n_{1239}$, $13n_{1280}$, $13n_{1281}$, $13n_{1282}$, $13n_{1283}$, $13n_{1284}$, $13n_{1285}$, $13n_{1286}$, $13n_{1287}$, $13n_{1288}$, $13n_{1289}$, $13n_{1290}$, $13n_{1291}$, $13n_{1292}$, $13n_{1293}$, $13n_{1294}$, $13n_{1295}$, $13n_{1305}$, $13n_{1306}$, $13n_{1307}$, $13n_{1308}$, $13n_{1309}$, $13n_{1310}$, $13n_{1311}$, $13n_{1312}$, $13n_{1313}$, $13n_{1314}$, $13n_{1315}$, $13n_{1316}$, $13n_{1317}$, $13n_{1318}$, $13n_{1319}$, $13n_{1320}$, $13n_{1321}$, $13n_{1322}$, $13n_{1323}$, $13n_{1324}$, $13n_{1332}$, $13n_{1333}$, $13n_{1334}$, $13n_{1363}$, $13n_{1377}$, $13n_{1378}$, $13n_{1379}$, $13n_{1380}$, $13n_{1381}$, $13n_{1385}$, $13n_{1386}$, $13n_{2933}$, $13n_{2934}$, $13n_{2935}$, $13n_{2936}$, $13n_{2937}$, $13n_{2954}$, $13n_{2955}$, $13n_{2956}$, $13n_{3390}$, $13n_{3505}$, $13n_{3723}$, $13n_{3725}$, $13n_{3726}$, $13n_{3727}$, $13n_{3763}$, $13n_{3838}$, $13n_{3915}$.

Appendix B. The knots with bridge index entry $[3, 4]$ that remain open

$13a_{129}$, $13a_{135}$, $13a_{136}$, $13a_{137}$, $13a_{194}$, $13a_{196}$, $13a_{197}$, $13a_{198}$, $13a_{233}$, $13a_{234}$, $13a_{239}$, $13a_{242}$, $13a_{243}$, $13a_{244}$, $13a_{253}$, $13a_{265}$, $13a_{268}$, $13a_{269}$, $13a_{270}$, $13a_{271}$, $13a_{272}$, $13a_{304}$, $13a_{309}$, $13a_{338}$, $13a_{362}$, $13a_{365}$, $13a_{366}$, $13a_{369}$, $13a_{373}$, $13a_{375}$, $13a_{387}$, $13a_{391}$, $13a_{392}$, $13a_{471}$, $13a_{483}$, $13a_{487}$, $13a_{516}$, $13a_{518}$, $13a_{530}$, $13a_{623}$, $13a_{629}$, $13a_{3145}$, $13a_{3147}$, $13a_{3148}$, $13a_{3149}$, $13a_{3167}$, $13a_{3168}$, $13a_{3169}$, $13a_{3170}$, $13a_{3188}$, $13a_{3189}$, $13a_{3190}$, $13a_{3191}$, $13a_{3204}$, $13a_{3205}$, $13a_{3207}$, $13a_{3208}$, $13a_{3209}$, $13a_{3210}$, $13a_{3213}$, $13a_{3216}$, $13a_{3219}$, $13a_{3220}$, $13a_{3221}$, $13a_{3222}$, $13a_{3223}$, $13a_{3232}$, $13a_{3234}$, $13a_{3248}$, $13a_{3259}$, $13a_{3261}$, $13a_{3262}$, $13a_{3263}$, $13a_{3264}$, $13a_{3266}$, $13a_{3274}$, $13a_{3275}$, $13a_{3276}$, $13a_{3291}$, $13a_{3292}$, $13a_{3293}$, $13a_{3294}$, $13a_{3295}$, $13a_{3298}$, $13a_{3301}$, $13a_{3302}$, $13a_{3303}$, $13a_{3317}$, $13a_{3320}$, $13a_{3323}$, $13a_{3330}$, $13a_{3331}$, $13a_{3335}$, $13a_{3345}$, $13a_{3346}$, $13a_{3347}$, $13a_{3348}$, $13a_{3351}$, $13a_{3354}$, $13a_{3355}$, $13a_{3356}$, $13a_{3363}$, $13a_{3365}$, $13a_{3368}$, $13a_{3374}$, $13a_{3375}$, $13a_{3388}$, $13a_{3390}$, $13a_{3393}$, $13a_{3394}$, $13a_{3395}$, $13a_{3396}$, $13a_{3399}$, $13a_{3400}$, $13a_{3401}$, $13a_{3402}$, $13a_{3403}$, $13a_{3404}$, $13a_{3405}$, $13a_{3407}$, $13a_{3409}$, $13a_{3410}$, $13a_{3411}$, $13a_{3412}$, $13a_{3413}$, $13a_{3415}$, $13a_{3417}$, $13a_{3418}$, $13a_{3811}$, $13a_{3812}$, $13n_{326}$, $13n_{329}$, $13n_{330}$, $13n_{341}$, $13n_{344}$, $13n_{345}$, $13n_{348}$, $13n_{351}$, $13n_{352}$, $13n_{355}$, $13n_{357}$, $13n_{358}$, $13n_{510}$, $13n_{513}$, $13n_{514}$, $13n_{518}$, $13n_{520}$, $13n_{521}$, $13n_{524}$, $13n_{525}$, $13n_{526}$, $13n_{529}$, $13n_{530}$, $13n_{531}$, $13n_{582}$, $13n_{590}$, $13n_{591}$, $13n_{598}$, $13n_{599}$, $13n_{600}$, $13n_{627}$, $13n_{632}$, $13n_{633}$, $13n_{644}$, $13n_{647}$, $13n_{648}$, $13n_{651}$, $13n_{652}$, $13n_{661}$, $13n_{662}$, $13n_{663}$, $13n_{694}$, $13n_{695}$, $13n_{696}$, $13n_{704}$, $13n_{705}$, $13n_{706}$, $13n_{707}$, $13n_{708}$, $13n_{709}$, $13n_{712}$, $13n_{713}$, $13n_{716}$, $13n_{717}$, $13n_{718}$, $13n_{834}$, $13n_{838}$, $13n_{839}$, $13n_{857}$, $13n_{926}$, $13n_{991}$, $13n_{992}$, $13n_{993}$, $13n_{1000}$, $13n_{1001}$, $13n_{1004}$, $13n_{1005}$, $13n_{1006}$, $13n_{1007}$, $13n_{1011}$, $13n_{1012}$, $13n_{1013}$, $13n_{1039}$, $13n_{1040}$, $13n_{1041}$, $13n_{1043}$, $13n_{1044}$, $13n_{1045}$, $13n_{1046}$, $13n_{1047}$, $13n_{1048}$, $13n_{1175}$, $13n_{1178}$, $13n_{1179}$, $13n_{1200}$, $13n_{1201}$, $13n_{1202}$, $13n_{1210}$, $13n_{1211}$, $13n_{1212}$, $13n_{1270}$, $13n_{1276}$, $13n_{1296}$,

$13n_{1367}$, $13n_{1370}$, $13n_{1371}$, $13n_{1374}$, $13n_{1375}$, $13n_{1376}$, $13n_{3739}$, $13n_{3742}$, $13n_{3745}$, $13n_{3748}$, $13n_{3770}$, $13n_{3773}$, $13n_{3776}$, $13n_{3779}$, $13n_{3814}$, $13n_{3840}$, $13n_{3842}$, $13n_{3844}$, $13n_{3854}$, $13n_{3855}$, $13n_{3856}$, $13n_{3870}$, $13n_{3871}$, $13n_{3872}$, $13n_{3873}$, $13n_{3874}$, $13n_{3897}$, $13n_{3898}$, $13n_{3899}$, $13n_{3900}$, $13n_{3901}$, $13n_{3903}$, $13n_{3904}$, $13n_{3905}$, $13n_{3911}$, $13n_{3914}$, $13n_{3916}$, $13n_{3918}$, $13n_{3919}$, $13n_{3920}$, $13n_{3921}$, $13n_{3922}$, $13n_{3924}$, $13n_{3925}$, $13n_{3928}$, $13n_{3929}$, $13n_{3931}$, $13n_{3932}$, $13n_{3933}$, $13n_{3934}$, $13n_{3935}$, $13n_{3936}$, $13n_{3937}$, $13n_{3938}$, $13n_{3939}$, $13n_{3940}$, $13n_{3941}$, $13n_{4587}$, $13n_{4589}$, $13n_{4633}$, $13n_{4634}$, $13n_{4637}$, $13n_{4638}$, $13n_{4639}$, $13n_{4640}$, $13n_{4641}$, $13n_{4642}$, $13n_{4643}$, $13n_{4644}$, $13n_{4645}$.

Appendix C. The knots of Theorem 1.2

$\gamma_4 = 1$ **(520 knots).**

12a_133	13a_6	13a_712	13a_1324	13a_2328	13a_3343
12a_200	13a_12	13a_714	13a_1362	13a_2339	13a_3354
12a_212	13a_45	13a_721	13a_1380	13a_2389	13a_3355
12a_218	13a_79	13a_726	13a_1412	13a_2402	13a_3356
12a_313	13a_129	13a_730	13a_1417	13a_2411	13a_3374
12a_335	13a_131	13a_732	13a_1421	13a_2414	13a_3399
12a_373	13a_136	13a_734	13a_1436	13a_2502	13a_3403
12a_390	13a_138	13a_776	13a_1438	13a_2505	13a_3408
12a_457	13a_156	13a_777	13a_1454	13a_2506	13a_3413
12a_474	13a_158	13a_786	13a_1477	13a_2511	13a_3415
12a_611	13a_166	13a_815	13a_1506	13a_2560	13a_3417
12a_639	13a_180	13a_823	13a_1511	13a_2595	13a_3418
12a_698	13a_197	13a_824	13a_1514	13a_2632	13a_3462
12a_709	13a_224	13a_827	13a_1523	13a_2645	13a_3463
12a_750	13a_232	13a_845	13a_1532	13a_2676	13a_3475
12a_818	13a_244	13a_861	13a_1598	13a_2788	13a_3477
12a_878	13a_247	13a_906	13a_1608	13a_2828	13a_3491
12a_882	13a_253	13a_916	13a_1609	13a_2864	13a_3494
12a_895	13a_265	13a_925	13a_1624	13a_2870	13a_3497
12a_916	13a_307	13a_988	13a_1627	13a_2893	13a_3505
12a_932	13a_318	13a_990	13a_1648	13a_2925	13a_3515
12a_1069	13a_338	13a_995	13a_1656	13a_3012	13a_3522
12a_1070	13a_342	13a_1000	13a_1658	13a_3026	13a_3556
12a_1092	13a_343	13a_1027	13a_1665	13a_3028	13a_3579
12a_1094	13a_353	13a_1068	13a_1666	13a_3039	13a_3598
12a_1102	13a_365	13a_1072	13a_1671	13a_3046	13a_3609
12a_1115	13a_369	13a_1077	13a_1673	13a_3057	13a_3611
12a_1206	13a_375	13a_1087	13a_1712	13a_3058	13a_3618
12a_1208	13a_381	13a_1103	13a_1719	13a_3076	13a_3653
12n_32	13a_392	13a_1106	13a_1722	13a_3167	13a_3657
12n_70	13a_396	13a_1122	13a_1727	13a_3181	13a_3674
12n_147	13a_410	13a_1131	13a_1731	13a_3188	13a_3689
12n_382	13a_476	13a_1134	13a_1756	13a_3190	13a_3692
12n_390	13a_505	13a_1140	13a_1758	13a_3198	13a_3741
12n_444	13a_518	13a_1163	13a_1767	13a_3242	13a_3758
12n_495	13a_519	13a_1169	13a_1851	13a_3258	13a_3764
12n_554	13a_530	13a_1176	13a_1856	13a_3266	13a_3772
12n_555	13a_537	13a_1186	13a_1862	13a_3274	13a_3776
12n_597	13a_538	13a_1193	13a_1907	13a_3275	13a_3786
12n_632	13a_550	13a_1230	13a_2059	13a_3279	13a_3787
12n_642	13a_591	13a_1231	13a_2094	13a_3294	13a_3799
12n_769	13a_610	13a_1237	13a_2166	13a_3298	13a_3805
12n_784	13a_619	13a_1250	13a_2218	13a_3302	13a_3807
12n_831	13a_680	13a_1277	13a_2308	13a_3306	13a_3811
12n_843	13a_687	13a_1316	13a_2311	13a_3328	13a_3866
13a_5	13a_699	13a_1320	13a_2323	13a_3333	13a_3905

13a_3928	13a_4544	13n_563	13n_1693	13n_3108	13n_4247
13a_3930	13a_4577	13n_620	13n_1726	13n_3190	13n_4272
13a_3955	13a_4591	13n_635	13n_1789	13n_3198	13n_4300
13a_3993	13a_4603	13n_671	13n_1932	13n_3236	13n_4305
13a_4054	13a_4620	13n_813	13n_1935	13n_3311	13n_4306
13a_4077	13a_4629	13n_816	13n_1976	13n_3334	13n_4313
13a_4103	13a_4647	13n_839	13n_1992	13n_3338	13n_4333
13a_4119	13a_4648	13n_852	13n_2021	13n_3344	13n_4334
13a_4141	13a_4654	13n_855	13n_2022	13n_3351	13n_4351
13a_4152	13a_4696	13n_892	13n_2080	13n_3402	13n_4354
13a_4178	13a_4697	13n_932	13n_2088	13n_3452	13n_4380
13a_4195	13a_4705	13n_988	13n_2115	13n_3501	13n_4457
13a_4198	13a_4743	13n_994	13n_2150	13n_3526	13n_4477
13a_4211	13a_4744	13n_999	13n_2193	13n_3602	13n_4485
13a_4220	13a_4746	13n_1005	13n_2245	13n_3641	13n_4509
13a_4247	13a_4794	13n_1057	13n_2265	13n_3667	13n_4515
13a_4249	13a_4848	13n_1061	13n_2383	13n_3695	13n_4517
13a_4268	13a_4874	13n_1062	13n_2404	13n_3723	13n_4521
13a_4289	13n_7	13n_1067	13n_2406	13n_3739	13n_4543
13a_4338	13n_30	13n_1111	13n_2409	13n_3740	13n_4614
13a_4342	13n_50	13n_1114	13n_2453	13n_3745	13n_4632
13a_4365	13n_54	13n_1201	13n_2503	13n_3752	13n_4642
13a_4367	13n_78	13n_1243	13n_2610	13n_3771	13n_4657
13a_4394	13n_149	13n_1275	13n_2629	13n_3776	13n_4666
13a_4398	13n_153	13n_1277	13n_2690	13n_3899	13n_4709
13a_4403	13n_157	13n_1302	13n_2694	13n_3902	13n_4746
13a_4412	13n_231	13n_1328	13n_2706	13n_3916	13n_4870
13a_4419	13n_290	13n_1419	13n_2741	13n_3919	13n_4896
13a_4426	13n_330	13n_1490	13n_2765	13n_3921	13n_4940
13a_4430	13n_341	13n_1492	13n_2773	13n_3923	13n_5006
13a_4437	13n_344	13n_1494	13n_2812	13n_3933	13n_5009
13a_4443	13n_347	13n_1500	13n_2834	13n_3968	13n_5012
13a_4449	13n_360	13n_1582	13n_2837	13n_4023	13n_5015
13a_4475	13n_370	13n_1585	13n_2851	13n_4030	13n_5041
13a_4479	13n_406	13n_1609	13n_2857	13n_4107	13n_5067
13a_4486	13n_418	13n_1628	13n_2926	13n_4115	13n_5089
13a_4487	13n_428	13n_1638	13n_2942	13n_4152	13n_5095
13a_4493	13n_520	13n_1670	13n_2949	13n_4155	13n_5104
13a_4511	13n_529	13n_1677	13n_2966	13n_4162	13n_5108
13a_4513	13n_530	13n_1678	13n_2977	13n_4193	
13a_4524	13n_561	13n_1688	13n_2985	13n_4202	

$\gamma_4 = 2$ **(1995 knots).**

11a_6	11a_163	11a_274	11a_329	12a_36	12a_88
11a_17	11a_173	11a_277	11a_337	12a_39	12a_96
11a_20	11a_178	11a_278	11a_338	12a_43	12a_97
11a_29	11a_188	11a_280	11a_339	12a_44	12a_103
11a_49	11a_192	11a_292	11a_352	12a_47	12a_113
11a_51	11a_203	11a_293	11a_354	12a_49	12a_114
11a_53	11a_206	11a_297	11a_355	12a_52	12a_117
11a_59	11a_214	11a_299	11a_360	12a_55	12a_134
11a_64	11a_240	11a_305	11a_362	12a_64	12a_136
11a_66	11a_241	11a_308	11a_366	12a_66	12a_137
11a_89	11a_248	11a_314	12a_9	12a_67	12a_143
11a_144	11a_259	11a_318	12a_29	12a_76	12a_146
11a_161	11a_264	11a_319	12a_35	12a_85	12a_148

12a_152	12a_628	12a_853	12a_1116	12n_260	13a_32
12a_156	12a_633	12a_855	12a_1120	12n_269	13a_33
12a_167	12a_636	12a_859	12a_1122	12n_289	13a_35
12a_176	12a_641	12a_869	12a_1137	12n_305	13a_40
12a_186	12a_647	12a_873	12a_1142	12n_317	13a_51
12a_188	12a_650	12a_876	12a_1143	12n_330	13a_52
12a_193	12a_651	12a_880	12a_1152	12n_361	13a_56
12a_195	12a_653	12a_881	12a_1159	12n_402	13a_58
12a_224	12a_655	12a_885	12a_1164	12n_432	13a_59
12a_239	12a_657	12a_890	12a_1171	12n_472	13a_65
12a_254	12a_661	12a_899	12a_1172	12n_503	13a_71
12a_264	12a_665	12a_905	12a_1177	12n_509	13a_82
12a_269	12a_669	12a_912	12a_1179	12n_517	13a_84
12a_276	12a_674	12a_923	12a_1180	12n_542	13a_85
12a_277	12a_681	12a_927	12a_1184	12n_557	13a_88
12a_289	12a_686	12a_937	12a_1189	12n_585	13a_89
12a_296	12a_689	12a_946	12a_1193	12n_603	13a_96
12a_297	12a_692	12a_950	12a_1197	12n_627	13a_97
12a_304	12a_693	12a_969	12a_1200	12n_638	13a_98
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13n_187	13n_755	13n_1240	13n_1933	13n_2722	13n_3333
13n_204	13n_767	13n_1273	13n_1936	13n_2751	13n_3337
13n_215	13n_771	13n_1278	13n_1950	13n_2775	13n_3354
13n_222	13n_773	13n_1294	13n_1998	13n_2805	13n_3368
13n_225	13n_776	13n_1300	13n_2042	13n_2816	13n_3419
13n_241	13n_787	13n_1323	13n_2061	13n_2827	13n_3426
13n_247	13n_832	13n_1329	13n_2065	13n_2870	13n_3427
13n_265	13n_841	13n_1332	13n_2089	13n_2881	13n_3432
13n_274	13n_858	13n_1335	13n_2104	13n_2883	13n_3435
13n_285	13n_895	13n_1341	13n_2110	13n_2898	13n_3447
13n_300	13n_900	13n_1344	13n_2114	13n_2899	13n_3454
13n_312	13n_907	13n_1382	13n_2121	13n_2900	13n_3461
13n_317	13n_914	13n_1383	13n_2123	13n_2901	13n_3470
13n_321	13n_916	13n_1390	13n_2125	13n_2907	13n_3476
13n_338	13n_929	13n_1406	13n_2131	13n_2920	13n_3484
13n_339	13n_935	13n_1415	13n_2143	13n_2929	13n_3486
13n_342	13n_943	13n_1417	13n_2159	13n_2935	13n_3491
13n_356	13n_947	13n_1421	13n_2171	13n_2939	13n_3493
13n_368	13n_961	13n_1439	13n_2190	13n_2959	13n_3497
13n_378	13n_962	13n_1498	13n_2192	13n_2972	13n_3499
13n_384	13n_965	13n_1503	13n_2195	13n_2981	13n_3502
13n_393	13n_974	13n_1504	13n_2231	13n_2995	13n_3508
13n_429	13n_981	13n_1508	13n_2248	13n_3000	13n_3515
13n_437	13n_983	13n_1516	13n_2252	13n_3003	13n_3518
13n_438	13n_1004	13n_1517	13n_2253	13n_3010	13n_3520
13n_458	13n_1008	13n_1519	13n_2280	13n_3015	13n_3532
13n_486	13n_1011	13n_1520	13n_2287	13n_3034	13n_3566
13n_495	13n_1029	13n_1528	13n_2297	13n_3038	13n_3576
13n_500	13n_1043	13n_1533	13n_2314	13n_3046	13n_3579
13n_503	13n_1049	13n_1535	13n_2335	13n_3048	13n_3583
13n_517	13n_1058	13n_1567	13n_2345	13n_3051	13n_3589
13n_519	13n_1063	13n_1590	13n_2346	13n_3060	13n_3593
13n_535	13n_1072	13n_1591	13n_2362	13n_3063	13n_3595
13n_546	13n_1075	13n_1592	13n_2368	13n_3068	13n_3608
13n_550	13n_1077	13n_1594	13n_2372	13n_3083	13n_3633
13n_569	13n_1080	13n_1595	13n_2421	13n_3103	13n_3643
13n_596	13n_1086	13n_1643	13n_2426	13n_3111	13n_3648
13n_601	13n_1095	13n_1654	13n_2437	13n_3113	13n_3653
13n_616	13n_1097	13n_1656	13n_2442	13n_3124	13n_3657

13n_3673	13n_3913	13n_4256	13n_4561	13n_4751	13n_4943
13n_3676	13n_3920	13n_4261	13n_4565	13n_4752	13n_4949
13n_3697	13n_3928	13n_4263	13n_4567	13n_4761	13n_4951
13n_3698	13n_3942	13n_4269	13n_4568	13n_4767	13n_4952
13n_3709	13n_3990	13n_4278	13n_4569	13n_4774	13n_4960
13n_3712	13n_4008	13n_4289	13n_4576	13n_4782	13n_4966
13n_3714	13n_4015	13n_4314	13n_4588	13n_4791	13n_4967
13n_3721	13n_4029	13n_4341	13n_4589	13n_4793	13n_4972
13n_3724	13n_4043	13n_4342	13n_4599	13n_4795	13n_4975
13n_3729	13n_4050	13n_4343	13n_4605	13n_4804	13n_4978
13n_3735	13n_4054	13n_4353	13n_4615	13n_4813	13n_4981
13n_3736	13n_4057	13n_4359	13n_4622	13n_4820	13n_4982
13n_3743	13n_4073	13n_4360	13n_4633	13n_4822	13n_4985
13n_3749	13n_4085	13n_4361	13n_4637	13n_4837	13n_4990
13n_3774	13n_4094	13n_4362	13n_4641	13n_4846	13n_4991
13n_3780	13n_4095	13n_4363	13n_4644	13n_4847	13n_4993
13n_3785	13n_4129	13n_4365	13n_4649	13n_4853	13n_4994
13n_3788	13n_4133	13n_4366	13n_4650	13n_4865	13n_4997
13n_3799	13n_4137	13n_4373	13n_4656	13n_4867	13n_5003
13n_3801	13n_4164	13n_4377	13n_4660	13n_4868	13n_5013
13n_3806	13n_4165	13n_4402	13n_4673	13n_4879	13n_5014
13n_3807	13n_4169	13n_4404	13n_4676	13n_4884	13n_5021
13n_3810	13n_4179	13n_4408	13n_4679	13n_4887	13n_5022
13n_3813	13n_4181	13n_4410	13n_4687	13n_4890	13n_5030
13n_3822	13n_4183	13n_4418	13n_4693	13n_4891	13n_5044
13n_3824	13n_4186	13n_4447	13n_4700	13n_4899	13n_5046
13n_3836	13n_4187	13n_4455	13n_4704	13n_4900	13n_5054
13n_3837	13n_4189	13n_4463	13n_4714	13n_4906	13n_5070
13n_3849	13n_4200	13n_4469	13n_4717	13n_4911	13n_5075
13n_3852	13n_4207	13n_4501	13n_4729	13n_4914	13n_5084
13n_3854	13n_4208	13n_4520	13n_4730	13n_4928	13n_5100
13n_3873	13n_4218	13n_4537	13n_4731	13n_4931	13n_5102
13n_3879	13n_4221	13n_4546	13n_4739	13n_4933	
13n_3882	13n_4223	13n_4550	13n_4743	13n_4936	
13n_3907	13n_4243	13n_4556	13n_4747	13n_4941	

$\gamma_4 = 3$ **(22 knots).**

12a_561	12a_906	13a_1557	13a_1992	13a_3585	13a_4637
12a_569	12a_907	13a_1662	13a_2890	13a_3747	13a_4674
12a_664	12a_1123	13a_1826	13a_3034	13a_3936	
12a_780	13a_1398	13a_1974	13a_3121	13a_4566	

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